

Mock Lab Test — Round 3: Dangerous Edition

Marking

Q1–Q8: Single answer → 1 mark each.

Q9–Q20: Select **ALL** correct answers.

For checkbox questions:

Correct selection → +1

Incorrect selection → -1

Missed correct answer → 0

Minimum per question → 0

Part A — Single Answer

Q1 — Train/Test Split Trap

A dataset has **500 samples**.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2
)
```

Assuming the split succeeds normally, how many samples are in X_train?

- A. 100
- B. 200
- C. 300
- D. 400

Q2 — Data Leakage Trap

Which is the **correct** sequence?

A.

Scale all data → Split train/test

B.

Split → Fit scaler on test data → Transform both datasets

C.

Split → Fit scaler on training data → Transform training and test data

D.

Split → Fit separate scalers on train and test data

Q3 — Correlation Trap

A correlation coefficient is:

$$r = 0.95$$

Which statement is most accurate?

- A. X definitely causes Y.
 - B. There is a strong positive linear relationship.
 - C. The relationship is perfectly positive.
 - D. There is no variation in the data.
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Q4 — PCA Trap

You apply PCA and keep enough components to explain **95% of the variance**.

What does this mean?

- A. 95% of the original rows are retained.
 - B. Exactly 95% of features are retained.
 - C. The selected components explain approximately 95% of the dataset's variance.
 - D. The model will have exactly 95% accuracy.
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Q5 — Gradient Descent Calculation

Given:

$$\theta = 5$$

$$\alpha = 0.2$$

$$\frac{\partial J}{\partial \theta} = -3$$

Calculate the new value: $\theta_{\text{new}} = \theta - \alpha \frac{\partial J}{\partial \theta}$

- A. 4.4
 - B. 5.6
 - C. 4.6
 - D. 5.4
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Q6 — Sigmoid Trap

If: $z = 0$

what is: $\sigma(z)$

- A. 0
 - B. 0.5
 - C. 1
 - D. Undefined
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Q7 — kNN Trap

Suppose the nearest neighbors have labels:

[1, 0, 1, 1, 0]

Using majority voting, what is the predicted class?

- A. 0
 - B. 1
 - C. 0.5
 - D. Cannot be determined
-

Q8 — argsort() Trap

Given:

distances = [5, 2, 8, 1]

What does this return?

np.argsort(distances)

- A. [1, 2, 5, 8]
 - B. [3, 1, 0, 2]
 - C. [5, 2, 8, 1]
 - D. [2, 1, 3, 0]
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Part B — Checkbox Questions

Q9 — StandardScaler vs MinMaxScaler

Select **ALL correct statements**.

- ☐ A. StandardScaler typically transforms features to approximately mean 0 and standard deviation 1.
 - ☐ B. MinMaxScaler typically maps values into a specified range such as 0 to 1.
 - ☐ C. StandardScaler automatically removes outliers.
 - ☐ D. Both scaling methods can be useful when numerical features have different scales.
 - ☐ E. MinMaxScaler is primarily used to balance class distributions.
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Q10 — Missing Values

Select **ALL correct statements**.

- ☐ A. df.isnull().sum() can be used to count missing values in each column.
 - ☐ B. Missing numerical values can sometimes be replaced with the mean.
 - ☐ C. Missing values should always be replaced with zero.
 - ☐ D. The appropriate treatment of missing values can depend on the dataset and feature.
 - ☐ E. dropna() can remove rows containing missing values.
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Q11 — Encoding Trap

Suppose:

Size = Small, Medium, Large

Select all correct statements.

- ☐ A. This variable has a natural order.
 - ☐ B. Ordinal encoding can be appropriate.
 - ☐ C. One-Hot Encoding is the only possible encoding method.
 - ☐ D. Assigning numerical values such as Small=0, Medium=1, Large=2 can represent the order.
 - ☐ E. StandardScaler is an encoding technique for categorical labels.
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Q12 — SMOTE / Oversampling / Undersampling

Select **ALL correct statements**.

- ☐ A. Random oversampling can increase the number of minority-class samples.
- ☐ B. Random undersampling can reduce the number of majority-class samples.
- ☐ C. SMOTE creates synthetic minority-class examples.

- ☐ D. SMOTE is a feature scaling technique.
 - ☐ E. Oversampling may increase the risk of overfitting because examples can be repeated.
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Q13 — Data Leakage: Dangerous Version

A student performs:

1. Split the data into train and test sets.
2. Fit StandardScaler using X_train.
3. Transform X_train.
4. Fit StandardScaler again using X_test.
5. Transform X_test.

Select all correct statements.

- ☐ A. Fitting the scaler on X_train is correct.
 - ☐ B. Transforming X_train using the training-fitted scaler is correct.
 - ☐ C. Fitting again on X_test is the recommended approach.
 - ☐ D. X_test should instead be transformed using the scaler fitted on X_train.
 - ☐ E. Training and test data should normally use the same learned scaling parameters.
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Q14 — Correlation Analysis

Select **ALL correct statements**.

- ☐ A. A correlation of +1 indicates a perfect positive linear relationship.
- ☐ B. A correlation of -1 indicates a perfect negative linear relationship.
- ☐ C. A correlation of 0 proves there is absolutely no relationship between two variables.
- ☐ D. A strong correlation proves causation.
- ☐ E. A scatter plot can help visualize the relationship between two numerical variables.

Q15 — PCA: Preserve Variance Trap

Select **ALL correct statements**.

- ☐ A. PCA can reduce the number of features.
 - ☐ B. Principal components are chosen based on variance.
 - ☐ C. Keeping more principal components can preserve more variance.
 - ☐ D. If PCA retains 90% explained variance, the resulting model must have 90% accuracy.
 - ☐ E. Scaling is often important before PCA when features have very different ranges.
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Q16 — Model Fit Scenarios

Select **ALL correct statements**.

- ☐ A. Training accuracy 55%, testing accuracy 53% may indicate underfitting.
 - ☐ B. Training accuracy 99%, testing accuracy 60% may indicate overfitting.
 - ☐ C. Training accuracy 90%, testing accuracy 88% may indicate a reasonably good fit.
 - ☐ D. A small difference between training and testing performance always proves the model is perfect.
 - ☐ E. A model with 100% training accuracy can never generalize well.
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Q17 — Logistic Regression vs Linear Regression

Select **ALL correct statements**.

- ☐ A. Logistic Regression is commonly used for classification.
- ☐ B. Linear Regression is commonly used to predict continuous numerical values.

- ☐ C. Logistic Regression commonly applies a sigmoid function to a linear combination.
 - ☐ D. Logistic Regression classifies points by finding the nearest K neighbors.
 - ☐ E. The expression $z = Xw + b$ can be used as the linear input before sigmoid in Logistic Regression.
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Q18 — Gradient Descent Traps

Select **ALL correct statements**.

- ☐ A. The gradient points in the direction of steepest increase of the cost function.
 - ☐ B. Gradient Descent generally moves in the opposite direction of the gradient.
 - ☐ C. A very small learning rate may cause slow convergence.
 - ☐ D. A very large learning rate may overshoot the minimum.
 - ☐ E. Gradient Descent always reaches the global minimum regardless of the problem.
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Q19 — kNN Traps

Select **ALL correct statements**.

- ☐ A. kNN can use majority voting for classification.
 - ☐ B. A smaller distance generally means two data points are closer.
 - ☐ C. A small K may be sensitive to noise.
 - ☐ D. A larger K always guarantees better accuracy.
 - ☐ E. Feature scaling can be important because kNN depends on distances.
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Q20 — Odd K + Multiple Classes Trap

Select **ALL correct statements**.

- ☐ A. An odd K can help reduce ties in binary classification.
 - ☐ B. K must always be odd.
 - ☐ C. Even values of K are valid values for kNN.
 - ☐ D. Odd K completely eliminates every possible tie in every multiclass classification problem.
 - ☐ E. Choosing K involves a trade-off between sensitivity to local patterns/noise and smoother predictions.
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Final Danger Question

Q21 — Select ALL Correct Answers

You have an imbalanced classification dataset and numerical features with very different ranges.

Which actions **may be appropriate**, depending on the algorithm and evaluation setup?

- ☐ A. Use a stratified split to help preserve class proportions.
- ☐ B. Fit a scaler on the entire dataset before splitting to make scaling more accurate.
- ☐ C. Fit preprocessing parameters using training data and apply them to test data.
- ☐ D. Consider SMOTE or another balancing technique on the training data.
- ☐ E. Apply SMOTE to the test set so the evaluation data becomes balanced.
- ☐ F. Consider feature scaling for distance-based methods such as kNN.
- ☐ G. Use the test data to help calculate preprocessing parameters because it contains useful information.